Dylan T. Jacobs
Curriculum Vitae

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Swarthmore College
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Education

Swarthmore College — Swarthmore, PA
B.S. General Engineering (electrical focus), B.A. Mathematics (applied math focus) — *2023–2027 (expected)*

GPA: 3.98/4.00

Relevant Coursework: Electronic Circuit Applications, Digital Signal Processing, Electrodynamics, Data Structures & Algorithms, Partial Differential Equations, Numerical PDEs

University of Carlos III, Madrid — Madrid, Spain
Study abroad in electrical engineering — *01/2026–05/2026*

Relevant Coursework: Computer Vision, Statistical Signal Processing

Areas of Interest

- Physics-informed neural networks (PINNs)
- Biomimetic and soft robots
- Control theory and reachability for autonomous systems
- Hyperbolic conservation laws and transport phenomena
- Kinetic modeling and Eulerian-Lagrangian methods

Publications

Pre-prints

[1] J. Nakao, **D.T. Jacobs**, and W. Taitano, "An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system", *Submitted* (2026).

In preparation

[1] J. Nakao, **D.T. Jacobs**, D.P. Hayes, and W. Guo, "An Eulerian-Lagrangian finite volume method with CUR adaptive rank (EL-CUR) for nonlinear convection-diffusion equations", *In preparation* (2026).

Scholarships and Awards

- Clearfield ASHE Scholarship, *American Society of Highway Engineers* — 04/2026
- Delaware Valley Engineers Undergraduate Scholarship, *Delaware Valley Engineers Society* — 02/2025

- Donna Prentice Memorial Scholarship, *American Society of Civil Engineers* 02/2024
- National Merit Scholarship 04/2023
- Global Seal of Biliteracy (Spanish) 07/2022

Internships and Research Experience

US Naval Research Laboratory — Washington, DC
Power Electronics Division — *05/2026–08/2026*

Mentors: Dr. Benjamin Greenberg and Dr. Boris Feigelson

Project 1: Developing inductively coupled plasma (ICP) reactor to synthesize nanoparticles and conducting ICP experiments.

- Minimized reflected power in ICP by tuning the RF matching network using Smith chart.
- Mitigated RF electromagnetic interference by designing Faraday cage.

Project 2: Scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDXS) of atomic-layer-deposition coated nanomaterials.

- Designed and ran atomic-layer-deposition (ALD) experiments to determine how dose parameters affect ALD coatings.
- Performed SEM and EDXS to characterize ALD-coated compacts and inform experimental design.

US Naval Research Laboratory — Washington, DC
Power Electronics Division — *05/2025–08/2025*

Mentor: Dr. Benjamin Greenberg

Project 1: Designing automated liquid-nitrogen (LN_2) delivery system to cool laser flash analysis (LFA) instrument.

- Designed and built Arduino-controlled LN2 dispenser with closed-loop thermocouple monitoring to automate LFA experiments, including a GUI and data-logging interface to the lab computer.

Project 2: Characterizing thermoelectric nanomaterials using laser flash analysis (LFA).

- Measured thermal diffusivity, specific heat, and Seebeck coefficients of thermoelectric semiconductors using LFA and Seebeck Analysis.
- Improved the measurement procedure for infrared-transparent and high-resistance samples.

Swarthmore College NaSCaRS Lab — Swarthmore, PA
Department of Mathematics and Statistics — *01/2024–present*

Mentor: Dr. Joseph Nakao

Project 1: Low-rank, structure-preserving solver for 1D2V Vlasov-Fokker-Planck plasma equation in cylindrical coordinates.

- Presented posters at 2025 SIAM NNP conference, 2024 Sigma Xi Swarthmore poster session.
- Giving talk at upcoming SIAM NNP 2026 conference.

Project 2: Eulerian-Lagrangian method with CUR decomposition for solving advection-diffusion equations in C++ and Python.

Swarthmore College Masroor Lab — Swarthmore, PA
Department of Engineering — *01/2024–05/2024*

Mentor: Dr. Emad Masroor

Project: Developing small-scale, oscillatory wind-energy harvesting device.

Oregon Health and Science University Portland, OR
Artificial Intelligence for Medical Systems Lab *06/2022–08/2022*

Mentor: Dr. Clara Mosquera-Lopez

Project: Coding Android app in Kotlin to detect heart murmers from Bluetooth audio data.

Conference Talks and Posters

Conference Talks

- An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system 10/2026 (upcoming)
  SIAM NNP Section Conference, Rutgers University, New Brunswick, New Brunswick, NJ

Posters

- An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system 02/2027 (upcoming)
  SIAM Conference on Computational Science and Engineering
- Plasma Synthesis and Scanning Electron Microscopy Characterization of Nanomaterials 09/2026
  Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA
- An adaptive and conservative low-rank IMEX solver for the hybrid ion Vlasov-Fokker-Planck and fluid electron system 11/2025
  SIAM NNP Section Conference, Penn State University, State College, PA
- Laser Flash Analysis Characterization of Thermoelectric Nanomaterials 09/2025
  Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA
- A Robust, Third-Order Accurate Solver for the 1D1V Vlasov-Poisson Equation 09/2024
  Sigma Xi Poster Session, Swarthmore College, Swarthmore, PA

Conferences Attended

- SIAM Conference Computational Science and Engineering 2026 02/2027 (upcoming)
  Pittsburgh, PA
- SIAM NNP Section Conference 2026 10/2026 (upcoming)
  Rutgers University, New Brunswick, NJ
- SIAM Conference on Computational Science and Engineering 2025 03/2025
  Fort Worth, TX
- SIAM NNP Section Conference 2025 11/2025
  Penn State University, State College, PA

Selected Projects

Convolutional Neural Network for Hand-Drawn Image Classification: link *05/2026*

*D*esigned and trained a CNN to classify 250 object categories, deployed it online for real-time evaluation using ONNX.

Component-Wise Kalman Filter Model for Smartphone Battery Drain: link
COMAP Mathematical Competition in Modeling (MCM) *01/2026*
Four-day team competition. Developed a battery state-space model with Kalman filtering using a system of ODEs, combined with Coulomb-Counting and battery Equivalent Circuit Model (ECM), to predict smartphone battery time-to-empty given time-varying operating and use conditions. Wrote and submitted a 25-page technical paper.

Analog Circuit and PCB Design *08/2025 —12/2025*

*D*esigned and fabricated PCBs in Fusion360 combining MOSFETs, sensors, actuators, op-amps, comparators, and audio amplifiers across four projects: analog audio equalizer, temperature-controlled fan, light-optimizing solar panel, regulated DC power supply.

AI Python Stock Trading Algorithm: link *02/2025*

*T*rained neural network on technical indicators to trade stocks using Alpaca API.

Generative Adversarial Network (GAN): link *02/2023*

*I*mage-generating model trained on web-scraped art.

Teaching Assistant Roles

Swarthmore College Swarthmore, PA
Mathematics Tutor *08/2025–present*

- Leading one-on-one tutoring sessions for Swarthmore mathematics students in a wide range of courses, from Calculus I, II, III to Differential Equations, to Real Analysis.

Swarthmore College Swarthmore, PA
Teaching Assistant for Electrical Circuit Analysis *08/2025–12/2025*

- Led tutoring, problem-solving, and review sessions for Swarthmore engineering students.
- Helped students develop problem-solving strategies, as well as foundations in mathematics, electrical engineering, and signal processing to solve circuit analysis problems.

Professional Involvement

Memberships

- National Society of Professional Engineers 09/2026–present
  Member.
- Tau Beta Pi Engineering Honor Society 02/2026–present
  Member of the Swarthmore College Chapter of the Tau Beta Pi Engineering Honor Society.
- Sigma Xi 08/2024–present
  Member of the Swarthmore College Chapter of Sigma Xi.

Extracurriculars

- Swarthmore Jazz Ensemble 08/2026–present
- Swarthmore Men's Varsity Soccer 08/2023–06/2026
- Swarthmore College Computer Society 01/2024–05-2024

Skills

Hardware: SPICE, PCB Design, CAD, Fusion360, Soldering, Embedded systems
Software: TensorFlow, PyTorch, OpenCV, Git, LaTeX
Programming: Python, Arduino, MATLAB, C/C++,
Languages: Spanish (fluent)